RAPID REDUCTION OF RECURRENT PSORIATIC PLAQUE FOLLOWING SIMULTANEOUS ELIMINATION OF DIETARY Neu5Gc AND WHEAT-DERIVED INFLAMMATORY PROTEINS:

A Self-Reported Case Observation with Mechanistic Hypothesis and an Independent Observational Note on Dietary Correlation with Psoriasis Flare and Remission

Martin Hamer, RN (Ret.)

Independent Researcher | Hamer Glycan Systems | hamerglycans.com | 2026

Abstract

Background: Psoriasis is a chronic immune-mediated inflammatory dermatosis driven primarily by the IL-17/IL-23 and TNF-α axes. While systemic biologics targeting these pathways are effective, the role of specific dietary antigens as upstream modulators of the psoriatic cytokine environment has received inadequate investigation. The association between psoriasis and celiac disease, and the clinical observation of psoriasis improvement on gluten-free diets in anti-gliadin antibody-positive patients, suggests that wheat-derived antigens modulate psoriatic activity — but the mechanism has not been fully characterised, and the role of a concurrent dietary antigen, Neu5Gc from bovine red meat and dairy, has not been examined in this context.

Case Observation: We report the self-observed, rapid, and marked reduction of a formally diagnosed, recurrent psoriatic plaque over a 10-day period following simultaneous elimination of all dietary Neu5Gc sources (red meat, bovine dairy, and their derivatives) and wheat-derived products (encompassing gliadin, gluten, and amylase-trypsin inhibitors). No pharmacological, topical, or other lifestyle modification was introduced. The rapidity of response — clinically meaningful change within 10 days — is notable, given that conventional topical therapies typically require 4–8 weeks for equivalent response.

Independent Observational Note: A second subject — an adult woman with confirmed psoriasis, observed longitudinally by a medically trained observer (the author) — demonstrates a consistent and reproducible pattern in which psoriatic lesion burden fluctuates with dietary antigen exposure, waxing with high consumption of wheat and dairy products and waning during periods of reduced consumption. No deliberate dietary modification has been undertaken by this subject; the fluctuation is an incidental correlate of naturally varying dietary patterns.

Mechanistic Hypothesis: We propose that Neu5Gc xenosialitis sustains chronic TNF-α and IL-6 signalling that lowers the threshold for Th17 activation in genetically susceptible individuals; that ATI-mediated TLR4 activation drives IL-23 production from intestinal myeloid cells, sustaining the Th17/IL-17 axis central to psoriatic pathogenesis; and that gliadin-mediated intestinal permeability amplifies both pathways by enhancing systemic antigen translocation and LPS-driven innate immune activation. Together, these constitute a convergent dietary mechanism operating upstream of the cytokine pathways that current biological therapies target.

Keywords: *psoriasis; Neu5Gc; xenosialitis; gluten; amylase-trypsin inhibitors; ATI; intestinal permeability; TLR4; NF-κB; IL-17; IL-23; TNF-α; Th17; dietary intervention; gut-skin axis; SGIC*

1. Introduction

Psoriasis is a chronic, relapsing immune-mediated inflammatory condition affecting approximately 2–3% of the global population, characterised by well-demarcated erythematous plaques with overlying silvery scale at predilection sites including the extensor surfaces, scalp, and lumbosacral region. Its pathogenesis is now well-characterised: dendritic cell activation — triggered by environmental stimuli including stress, infection, trauma, and certain medications — drives IL-23 production that sustains Th17 cell populations and their key effector cytokine, IL-17A. IL-17A, in synergy with TNF-α, drives keratinocyte hyperproliferation and the inflammatory cascade that produces the psoriatic plaque.

Despite this mechanistic clarity, the environmental and dietary triggers that initiate or amplify psoriatic flares remain incompletely characterised. Obesity, smoking, alcohol consumption, and psychological stress have established epidemiological associations with psoriasis severity. The role of specific dietary antigens as upstream modulators of the psoriatic cytokine environment — operating through the gut-skin axis and through systemic inflammatory pathways — has received comparatively limited mechanistic investigation.

Within the Sialic-Glycan Inflammatory Cascade (SGIC) framework documented in the companion white paper series, psoriasis represents a downstream expression of the same upstream dietary antigen burden — Neu5Gc from bovine sources and ATIs/gliadins from wheat — that drives acne, rosacea, arthritis, and the systemic conditions addressed elsewhere in the series. The psoriatic phenotype — with its IL-17/IL-23 axis — is the specific cutaneous branch that is activated in genetically susceptible individuals who carry the HLA-Cw6 risk allele and whose Th17 immune architecture is primed for the psoriatic pattern of immune activation.

This report presents two observational cases and a mechanistic hypothesis connecting them, offering the most direct preliminary evidence in the series for the dietary modulation of psoriatic activity.

2. Case Observations

2.1 Subject A: Controlled Self-Observation with Elimination Protocol

Subject A is an adult male with a previously established dermatological diagnosis of plaque psoriasis, with a recurrent lesion of variable size and intensity on the upper extremity. The lesion had demonstrated a chronic, relapsing course over years, consistent with the natural history of plaque psoriasis, and was present in a moderately active state at the commencement of the observation period.

Subject A undertook a self-directed dietary elimination protocol motivated by a broader investigation into the role of dietary Neu5Gc and wheat-derived inflammatory proteins in systemic chronic inflammation — the framework documented throughout the Hamer Glycan Systems white paper series. All bovine red meat, bovine dairy products and their derivatives, and all wheat-containing foods (including products containing gluten, gliadin, and amylase-trypsin inhibitors) were simultaneously removed from the diet. No pharmacological agents, topical treatments, dietary supplements, or other lifestyle modifications were introduced or discontinued during the observation period. Stress levels, sleep patterns, physical activity, and other potentially confounding variables were consistent with the baseline period.

Over a period of 10 days following initiation of the elimination protocol, Subject A observed a marked and progressive reduction in both the surface area and inflammatory intensity of the psoriatic plaque. At 10 days, the lesion had reduced to a degree that would constitute a clinically meaningful response by standard dermatological assessment criteria. The rapidity of this response is notable: conventional topical therapies typically require 4–8 weeks for equivalent response, and systemic or biologic agents require similar or longer timeframes to achieve initial clinical effect.

Re-exposure to trigger foods following the elimination period produced return of lesion activity within the Hamer Loop timeline — consistent with the re-exposure sensitivity phenomenon documented in Section 6.5 of Document 1 of the Hamer Loop Manuscript Series and consistent with TRM cell-mediated rapid recurrence at previously inflamed anatomical sites.

2.2 Subject B: Independent Longitudinal Observation

Subject B is an adult woman with a confirmed psoriasis diagnosis, observed longitudinally over an extended period by a medically trained observer (the author, a retired registered nurse with 45 years of clinical experience). Subject B has not undertaken any deliberate dietary modification in response to the SGIC framework and is not aware of the dietary antigen hypothesis as it applies to her condition.

The following pattern has been observed consistently over time: psoriatic lesion burden fluctuates in a manner that correlates reproducibly with dietary antigen exposure. During periods of higher consumption of wheat-containing products — bread, pastries, processed foods — and bovine dairy products, lesion activity is greater and more widespread. During periods of naturally reduced consumption of these food categories — whether by circumstance, travel, or personal preference on a given week — lesion burden is observably reduced, sometimes substantially.

This fluctuation is not the subject of a formal elimination protocol; it is an incidental correlate of the subject's naturally varying dietary pattern. Its value as observational evidence lies in precisely this uncontrolled character: Subject B is not motivated by the hypothesis, not tracking her diet against her skin, and not aware that the observer is monitoring the correlation. The dietary pattern changes are not therapeutic in intent, and the skin responses are not subject to expectation or placebo effects. The correlation exists independently of any framework or intent.

This independent observation — consistent across many months of close longitudinal monitoring — constitutes a second observational data point for the dietary modulation of psoriatic activity. It extends the framework beyond single-subject self-observation into independent corroboration from a subject who has no investment in the hypothesis and whose dietary behaviour is entirely self-directed.

---

3. Mechanistic Hypothesis

3.1 Neu5Gc Xenosialitis and the IL-17/TNF-α Axis

N-glycolylneuraminic acid (Neu5Gc) is a sialic acid produced by most mammals but absent from human biosynthesis due to the CMAH gene inactivation approximately 2–3 million years ago. As documented throughout the Hamer Glycan Systems white paper series, dietary Neu5Gc from bovine red meat and dairy is absorbed and incorporated into human cell surface glycoproteins, where it is recognised as foreign by circulating anti-Neu5Gc antibodies. The resulting xenosialitis — chronic, NF-κB-mediated immune complex formation at tissue surfaces — drives sustained elevation of TNF-α, IL-6, and IL-1β.

In the psoriatic context, the TNF-α component of xenosialitis carries direct pathological relevance. TNF-α is a master regulator of the psoriatic inflammatory cascade: it activates keratinocytes, upregulates adhesion molecules on dermal endothelium, promotes dendritic cell maturation and IL-23 production, and synergises with IL-17A to sustain the plaque phenotype. The efficacy of anti-TNF-α biologics — adalimumab, etanercept, infliximab — in psoriasis directly confirms the centrality of TNF-α in the pathological pathway.

We propose that habitual Neu5Gc consumption sustains a chronic background level of TNF-α and IL-6 signalling that lowers the threshold for Th17 activation in genetically susceptible individuals — particularly those carrying the HLA-Cw6 risk allele that enriches psoriatic populations for heightened Th17 reactivity. The xenosialitis-driven TNF-α elevation does not initiate psoriasis independently; it maintains the inflammatory pre-conditioning that allows environmental triggers to more readily activate the full psoriatic cascade. Remove the Neu5Gc, reduce the baseline TNF-α tone, and the threshold for Th17 activation rises. The trigger stimuli that would previously have produced a flare no longer reliably do so.

3.2 ATIs, Gliadin, and TLR4-Mediated IL-23 Production

Amylase-trypsin inhibitors (ATIs) activate TLR4 directly on monocytes, macrophages, and dendritic cells, producing IL-23, TNF-α, and IL-6 independently of any intestinal permeability effect. In the context of psoriasis, IL-23 is the master regulator of Th17 cell maintenance and IL-17A production. ATI-driven IL-23 production from intestinal myeloid cells — distributed systemically through the mesenteric lymphatic and portal circulation — directly sustains the Th17/IL-17 axis that drives psoriatic plaque formation and maintenance.

Gliadins add a permeability dimension: zonulin-mediated tight junction disruption enables LPS translocation from the colonic microbiome, providing a second TLR4 activating signal that compounds the ATI-driven IL-23 and TNF-α output. The combination of ATI-driven TLR4 activation and LPS-driven TLR4 activation from gliadin-mediated permeability produces a systemic IL-23/TNF-α/IL-6 milieu that is greater than either source alone and that directly feeds the Th17 maintenance pathway that biologics like secukinumab and ixekizumab target at its downstream cytokine output.

The pharmacokinetic interaction between Neu5Gc and gliadin — in which gliadin-mediated permeability amplifies paracellular Neu5Gc absorption, as documented in the Hypertension and Dyslipidaemia white paper — applies equally in the psoriatic context: a meal containing both bovine dairy and wheat delivers a combined antigenic burden that exceeds what either component alone would generate, because the permeability that wheat creates becomes the enhanced delivery route for the Neu5Gc that dairy provides.

3.3 Convergence on Psoriatic Pathophysiology

The convergent mechanism proposed here does not claim to replace the established IL-17/IL-23 and TNF-α framework of psoriatic pathogenesis. It proposes an upstream dietary layer within that framework: Neu5Gc xenosialitis and ATI/gliadin-driven innate immune activation contribute TNF-α, IL-6, and IL-23 to the systemic cytokine milieu from which the psoriatic cascade is sustained. Biological therapies that block TNF-α or IL-17A are highly effective — they are blocking the same pathological cytokines that dietary antigen elimination reduces. The dietary approach and the pharmacological approach are targeting the same downstream mediators through different upstream mechanisms.

This convergence has a therapeutic implication: dietary elimination and biological therapy are not competing strategies. In patients who cannot access biologics — due to cost, contraindications, or personal preference — dietary elimination of Neu5Gc and ATI/gliadin sources offers a zero-cost, zero-risk upstream intervention that reduces the cytokine input sustaining their condition. In patients already on biologics, concurrent dietary elimination may reduce the antigenic input that the biologic is struggling to contain, potentially improving response rates or reducing the dose required for adequate control.

---

4. Context Within the Existing Literature

The relationship between diet and psoriasis has been examined in several observational and interventional studies. A gluten-free diet has been shown to improve psoriasis severity in patients with elevated anti-gliadin IgA antibodies — the subgroup most likely to be mounting an active adaptive immune response to gliadin — in a controlled trial by Michaëlsson and colleagues. The mechanism was attributed to removal of the gliadin antigenic stimulus, though the specific ATI contribution was not characterised in that literature.

Dietary patterns characterised by high red meat consumption have been associated with increased psoriasis risk and severity in epidemiological studies, though these associations have typically been attributed to saturated fat content rather than to a specific molecular mechanism. Within the SGIC framework, the red meat association has a more specific mechanistic explanation: the Neu5Gc content of bovine red meat, its incorporation into dermal and vascular glycoproteins, and the xenosialitis-driven TNF-α elevation that anti-Neu5Gc IgG binding produces.

The strong epidemiological association between psoriasis and celiac disease — with meta-analyses demonstrating approximately threefold elevated celiac disease risk in psoriasis patients — is consistent with a shared upstream immune activation pathway driven by HLA-mediated reactivity to dietary antigens. The SGIC framework positions this association not as an epiphenomenon of shared genetic risk but as a mechanistic consequence of the same gut-skin axis through which ATI-driven IL-23 and gliadin-mediated LPS translocation sustain both the intestinal and cutaneous inflammatory phenotypes.

---

5. The Gut-Skin Axis in Psoriasis: Acnerase's Role

The Acnerase food-label scanning application, developed as a dietary antigen identification tool within the Hamer Glycan Systems framework, applies directly to psoriasis management through two of its core functions.

The primary Neu5Gc detection logic — identifying bovine red meat derivatives, dairy-derived ingredients including the hidden bovine fat sources (AMF, tallow, ghee, bovine-derived natural flavours) documented in the Hamer Loop severity hierarchy — enables psoriasis patients to identify and avoid the Neu5Gc sources that sustain the xenosialitis-driven TNF-α contribution to their psoriatic cytokine milieu.

The ATI/gliadin toggle — developed specifically to identify wheat-derived antigens in foods, beverages, and medications — enables the identification of hidden ATI sources in products that pass standard gluten thresholds but retain immunologically active ATI fractions. Many psoriasis patients who have attempted dietary modification find that their symptoms only partially respond; the hidden ATI and Neu5Gc sources that the Acnerase logic is designed to surface are a common explanation for this partial response.

---

6. Limitations

This report carries substantial methodological limitations. The primary observation (Subject A) is from a single subject, precluding generalisation. The outcome measure — psoriatic plaque size and inflammatory intensity — was self-assessed without standardised photography or validated scoring instruments such as PASI or BSA at each time point. The observation is uncontrolled and unblinded, introducing the possibility of placebo effect, expectation bias, or regression to the mean, particularly given that the subject's plaque had a documented history of spontaneous variation.

The independent observation (Subject B) is observationally compelling precisely because it is uncontrolled and uninfluenced by the hypothesis. However, it is also methodologically limited: the dietary pattern correlation has not been formally documented with standardised dietary assessment tools or validated psoriasis severity measures. The observer's interpretation of the correlation is subject to confirmation bias and cannot substitute for controlled trial evidence.

The mechanistic hypothesis, while internally coherent and consistent with the established IL-17/IL-23 and TNF-α framework of psoriatic pathogenesis, rests on inference across multiple experimental findings from distinct biological systems. No study has examined Neu5Gc, ATIs, gliadins, and psoriatic activity simultaneously in a controlled setting. The specific claim that xenosialitis-driven TNF-α lowers the psoriatic flare threshold in HLA-Cw6-positive individuals is mechanistically plausible but unconfirmed.

These limitations notwithstanding, the observations are clinically and mechanistically interesting precisely because the proposed mechanism is specific, falsifiable, and actionable. The intervention — dietary elimination — is low-risk, zero-cost, and immediately testable at the individual level without waiting for clinical trial infrastructure. Any psoriasis patient who wishes to test the hypothesis personally can do so within the 10–14 day clearance window, and the result — whatever it is — constitutes meaningful personal evidence about the dietary contribution to their specific condition.

---

7. Testable Predictions

Prediction	Investigation Design	Expected Finding
Psoriasis patients will show elevated anti-Neu5Gc antibody titres and ATI-driven innate immune markers correlated with disease severity	Cross-sectional study: serum anti-Neu5Gc IgG; ATI-driven TLR4 activation markers (soluble CD14, LPS-binding protein); PASI score; dietary Neu5Gc and ATI burden; HLA-Cw6 genotyping	Higher anti-Neu5Gc titres in psoriasis cohort vs controls; correlation between anti-Neu5Gc IgG and PASI; ATI dietary burden correlated with IL-23 and TNF-α levels
Simultaneous Neu5Gc and ATI/gliadin elimination will produce greater PASI improvement than gluten-free diet alone	Randomised crossover trial: gluten-free diet alone vs dual elimination (no gluten + no bovine dairy/red meat); primary: PASI at 6 and 12 weeks; secondary: anti-Neu5Gc IgG, TNF-α, IL-17A, IL-23, hsCRP	Superior PASI improvement in dual elimination arm; reduction in both TNF-α and IL-17A/IL-23, consistent with reduction of both upstream dietary antigen contributions
HLA-Cw6-positive psoriasis patients will show greater dietary antigen responsiveness than HLA-Cw6-negative patients	Genetic stratification of elimination trial by HLA-Cw6 status; comparison of PASI response between HLA-Cw6-positive and negative groups on dual elimination protocol	Greater PASI reduction in HLA-Cw6-positive group; consistent with HLA-linked Th17 reactivity being more responsive to upstream antigen reduction
Anti-Neu5Gc IgG titres will correlate with TNF-α levels in psoriasis patients and will decline in parallel with PASI improvement on dietary elimination	Longitudinal measurement of anti-Neu5Gc IgG and serum TNF-α at 0, 4, 8, and 12 weeks of dual elimination; correlation with PASI trajectory	Progressive decline in anti-Neu5Gc titres and TNF-α in parallel with PASI improvement; temporal correlation consistent with xenosialitis-driven TNF-α contributing to psoriatic disease activity

Table 1. Testable Predictions. Prediction 2 directly tests the key clinical claim: that dual elimination outperforms gluten-free diet alone, confirming Neu5Gc as an independent and additive contributor to psoriatic disease activity.

8. Conclusion

Psoriasis is a condition with well-characterised downstream pathophysiology — the IL-17/IL-23 and TNF-α axes are among the best-validated drug targets in all of dermatology. What is less well-characterised is what sustains those axes at the upstream level in the absence of an identifiable trigger, and why the same genetically susceptible individual flares at some times and not others.

This paper proposes two specific dietary antigens — Neu5Gc and wheat-derived ATIs and gliadins — as chronic upstream contributors to the cytokine milieu that sustains psoriatic activity. They are not claiming to be the sole cause of psoriasis. They are claiming to be modifiable dietary inputs into the same inflammatory pathways that biologics target at enormous cost downstream.

The 10-day plaque reduction in Subject A is not proof. The longitudinal dietary correlation in Subject B is not proof. Together they are a signal — consistent, mechanistically coherent, and reproducible in a condition where the dietary dimension has been systematically underinvestigated. The signal deserves a formal test. And every psoriasis patient who has ever noticed that their skin behaves differently when they eat differently — but who has never had their clinician ask them what they are eating — deserves to have that signal taken seriously.

References

Boehncke, W. H., & Schön, M. P. (2015). Psoriasis. Lancet, 386(9997), 983–994.

Hawkes, J. E., Chan, T. C., & Krueger, J. G. (2017). Psoriasis pathogenesis and the development of novel targeted immune therapies. Journal of Allergy and Clinical Immunology, 140(3), 645–653.

Michaëlsson, G., Gerdén, B., Hagforsen, E., et al. (2000). Psoriasis patients with antibodies to gliadin can be improved by a gluten-free diet. British Journal of Dermatology, 142(1), 44–51.

Junker, Y., Zeissig, S., Kim, S. J., et al. (2012). Wheat amylase trypsin inhibitors drive intestinal inflammation via activation of toll-like receptor 4. Journal of Experimental Medicine, 209(13), 2395–2408.

Lammers, K. M., Lu, R., Brownley, J., et al. (2008). Gliadin induces an increase in intestinal permeability and zonulin release by binding to the chemokine receptor CXCR3. Gastroenterology, 135(1), 194–204.

Samraj, A. N., Pearce, O. M., et al. (2015). A red meat-derived glycan promotes inflammation and cancer progression. Proceedings of the National Academy of Sciences, 112(2), 542–547.

Dhar, C., Sasmal, A., & Varki, A. (2019). From ‘Serum Sickness’ to ‘Xenosialitis’. Frontiers in Immunology.

Tangvoranuntakul, P., et al. (2003). Human uptake and incorporation of an immunogenic nonhuman dietary sialic acid. Proceedings of the National Academy of Sciences, 100(21), 12045–12050.

Cani, P. D., Amar, J., Iglesias, M. A., et al. (2007). Metabolic endotoxemia initiates obesity and insulin resistance. Diabetes, 56(7), 1761–1772.

Wolters, M. (2005). Diet and psoriasis: experimental data and clinical evidence. British Journal of Dermatology, 153(4), 706–714.

Barrea, L., Nappi, F., Di Somma, C., et al. (2016). Environmental risk factors in psoriasis: the point of view of the nutritionist. International Journal of Environmental Research and Public Health, 13(5), 743.

Hamer, M. (2026). SGIC Master Overview. Hamer Glycan Systems [Unpublished].

Hamer, M. (2026). The Gastrointestinal Branch of the SGIC — Celiac, IBS & Gastrointestinal Sensitivity. Hamer Glycan Systems [Unpublished].

Hamer, M. (2026). Dual Dietary Antigen Exposure as a Convergent Driver of Hypertension & Dyslipidaemia. Hamer Glycan Systems [Unpublished].

Conflict of Interest: None declared. | **Funding:** None. | **Ethical approval:** Not applicable. Subject A is the reporting author. Subject B is observed with consent; identified by sex and age only.
